Informatics for Chemistry, Biology, and Biomedical Sciences & Rationality over fashion and hype in drug design

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- Informatics for Chemistry, Biology, and Biomedical Sciences. Edgar López-López, Jürgen Bajorath, José L. Medina-Franco; J. Chem. Inf. Model. 25 January 2021; 61 (1): 26–35.
- Medina-Franco JL, Martinez-Mayorga K, Fernández-de Gortari E et al. Rationality over fashion and hype in drug design. F1000Research 2021, 10(Chem Inf Sci):397.

Content

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Informatics for Chemistry, Biology, and Biomedical Sciences

- Informatics in the literature
- Some informatics disciplines
- A comparative analysis
- Bibliometrics
- Dissemination

Rationality over fashion and hype in drug design

- The “black box” AI problem
- Misconceptions vs Intended use
- Bridging the gap

Reflections

Informatics in the literature

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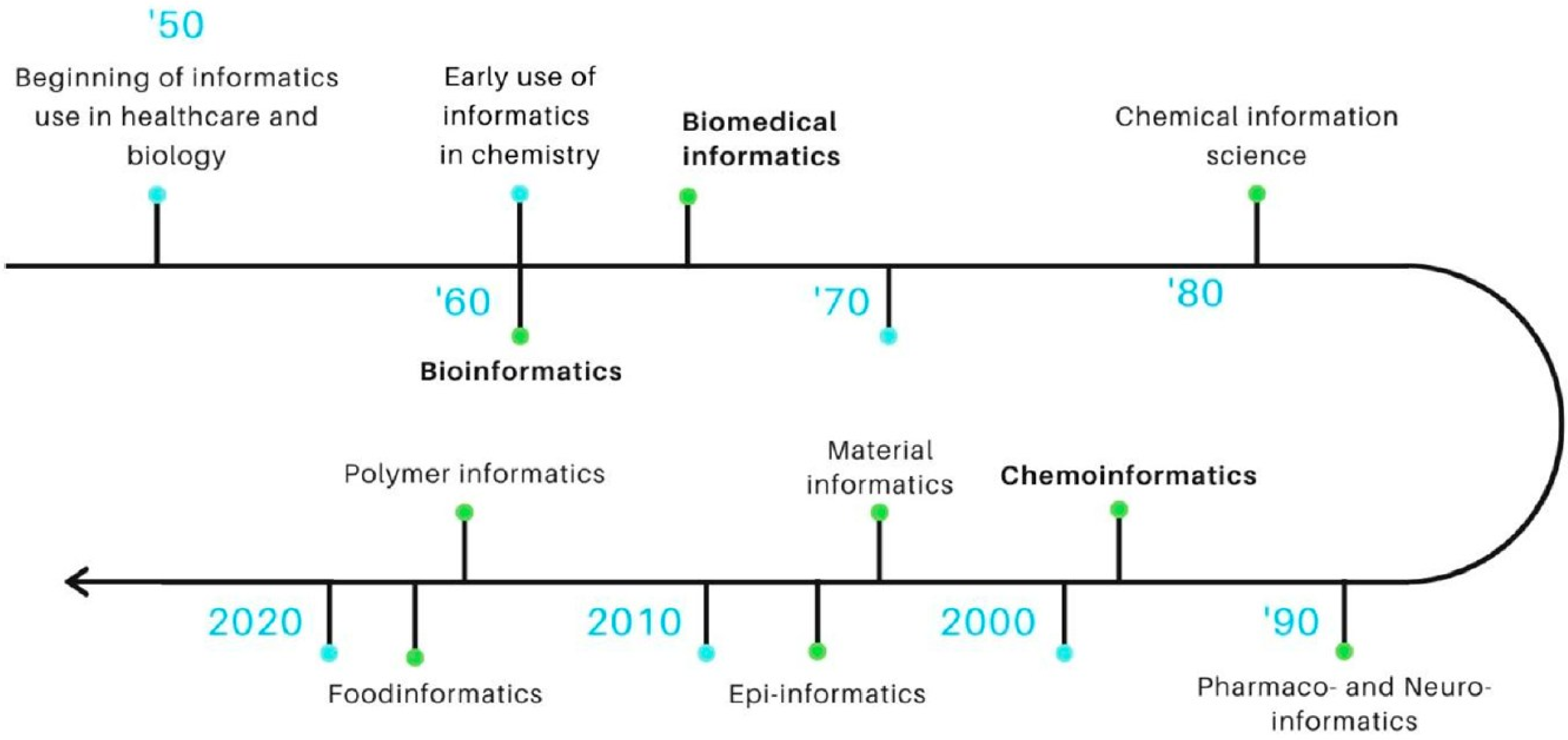

"The study of processes for storing and obtaining information"

Informatics disciplines in natural and life science

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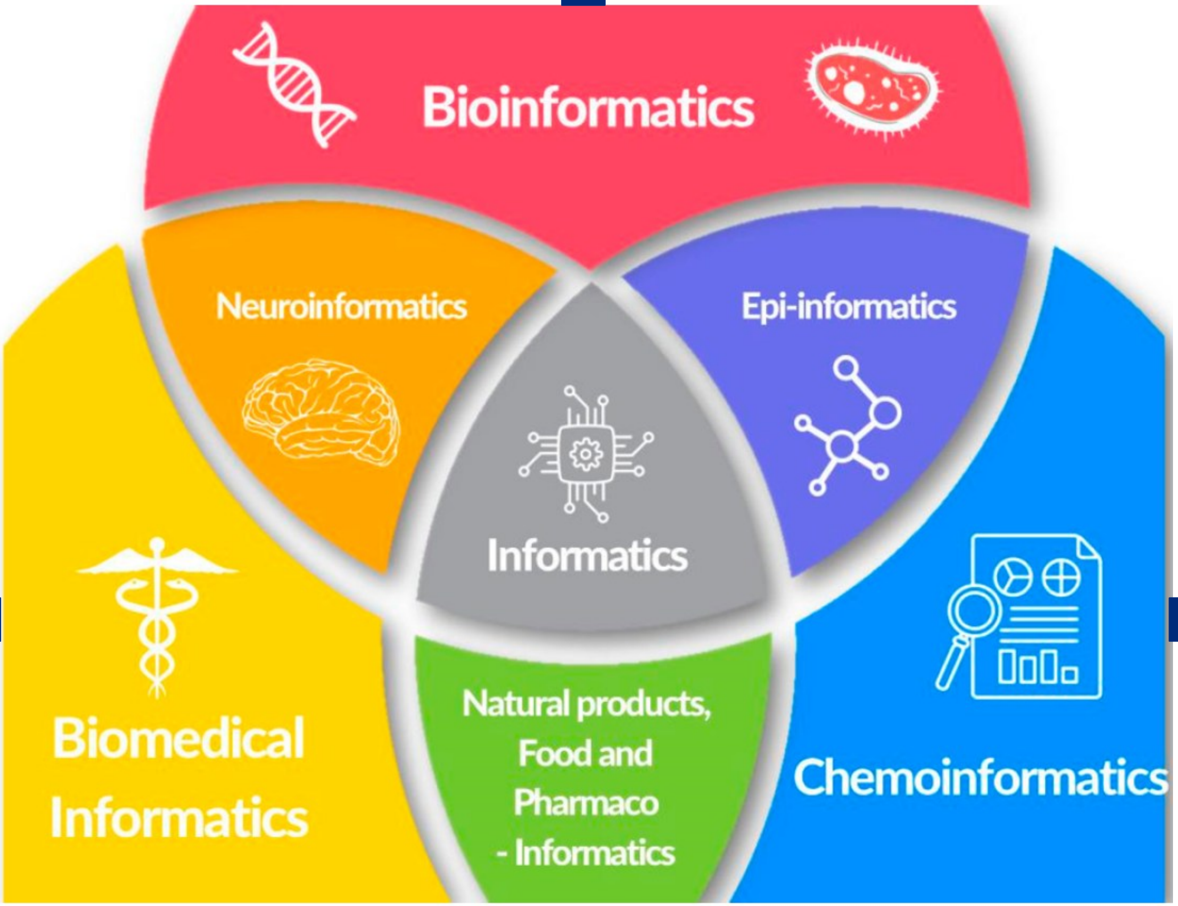
“Research field in which computer scientists, biologists, physicians, mathematicians, and chemists combine their expertise to collaborate on diverse tasks, from the discovery of new facts on complex biological systems to the rationalization of organization of health systems.”

“It is the application of the science of information as data plus meaning to problems of biomedical interest.”

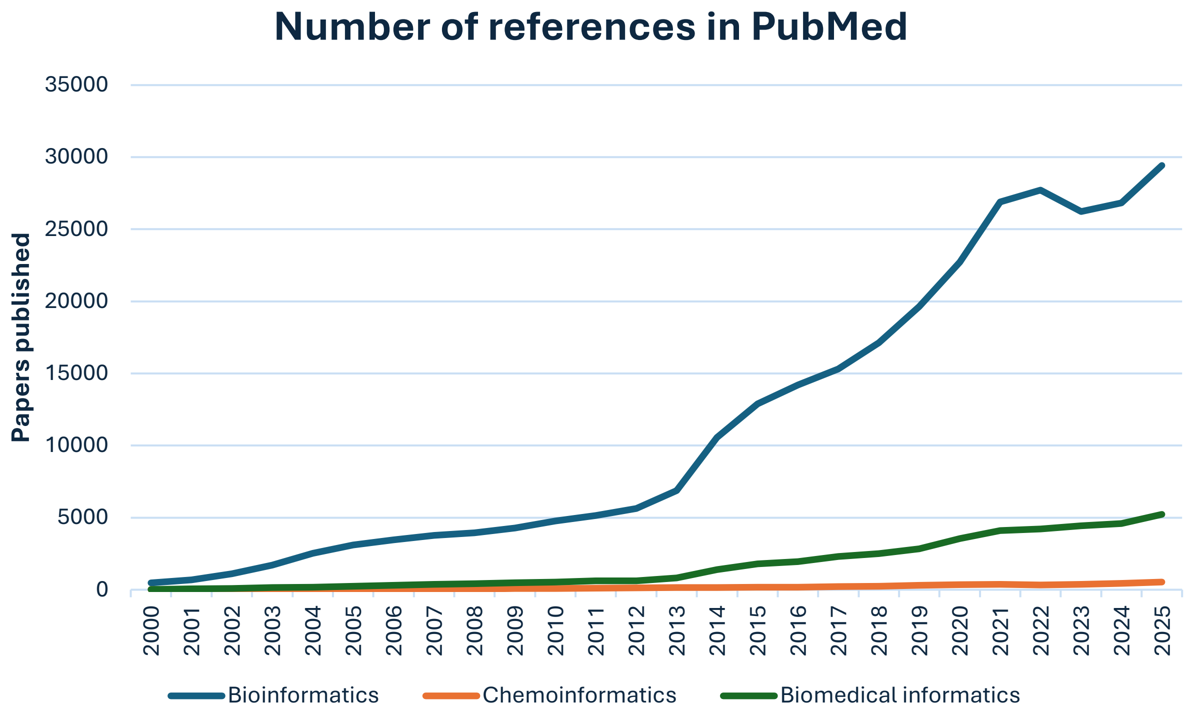


“Encompasses the design, creation, organization, management, retrieval, analysis, dissemination, visualization, and use of chemical information.”

A comparative analysis

	Chemoinformatics	**Bioinformatics**	**Biomedical informatics**
Main objective and scope; type of data	Small molecules and property annotations	Macromolecules and genes	Health-related data knowledge
Core identity	Molecular fingerprints	Sequences	Clinical/nonclinical patterns
Key concepts and techniques	Chemical space Similarity	Homology Aligment	Image and docs análysis
Examples of study	Property prediction Chemical libraries	Biomolecular databases Sequence análisis	Prediction of clinical diagnostic

Growth over the time



Keywords: "bioinformatics“, ("chemoinformatics" OR "cheminformatics"), "biomedical informatics“. 2000-2025

Bioinformatics

BMC Bioinformatics

Journal of Cheminformatics

JCIM JOURNAL OF CHEMICAL INFORMATION AND MODELING

Journal of Biomedical Informatics

International Journal of medicalinformatics

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Teaching and Dissemination

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Discipline ⬇️	**Chemoinformatics**	**Bioinformatics**	**Biomedical informatics**
Undergraduate programs	30	500	30
Master programs	10	450	120
Textbooks	17/100	364/300	22/80
Specialized journals	3	25	2

The “black box” AI problem

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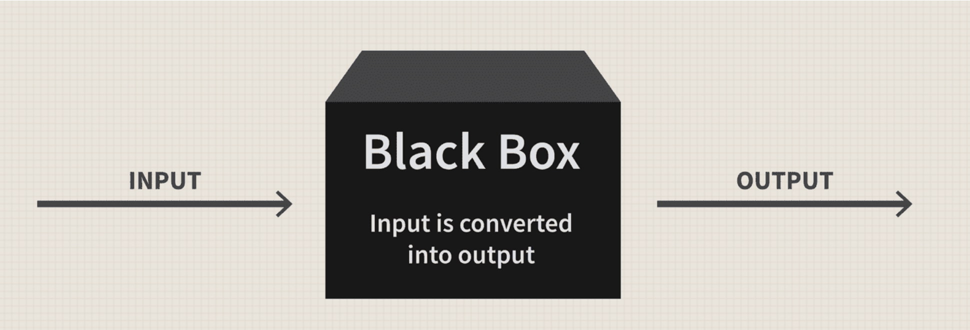
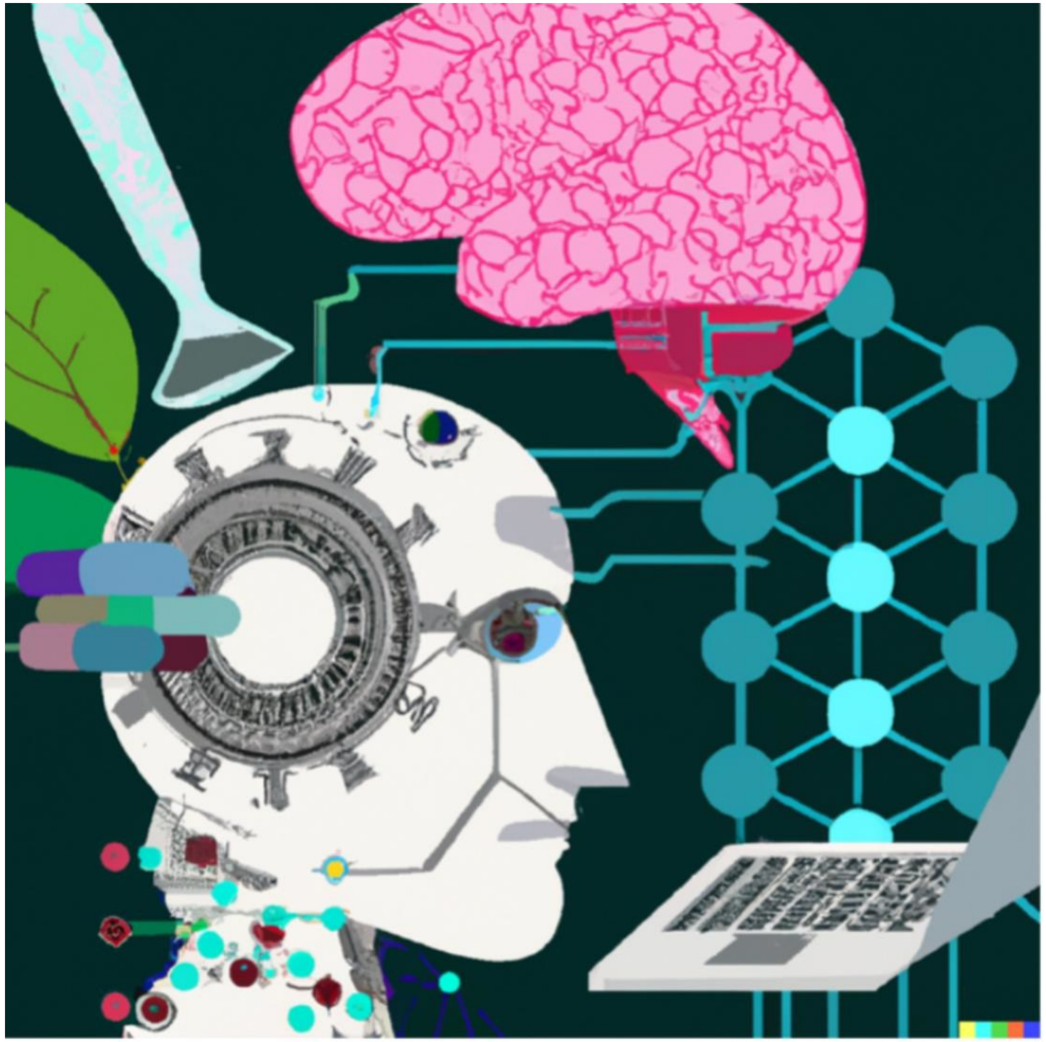


One of the critical limitations of current AI systems is the “black box” problem, where model predictions are difficult to interpret.



- Blanco-González A, Cabezón A, Seco-González A, Conde-Torres D, Antelo-Riveiro P, Piñeiro Á, Garcia-Fandino R. The Role of AI in Drug Discovery: Challenges, Opportunities, and Strategies. Pharmaceuticals. 2023; 16(6):891

Misconceptions vs Intended use

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Misconceptions	Correct meaning
"Machine learning and AI are the new standard for CADD".	Machine learning is a part of AI and already has a long record of use in CADD.
Molecular modeling and chemoinformatics are other terms for CADD.	CADD covers various theoretical disciplines including, molecular modeling, chemoinformatics, bioinformatics, theoretical chemistry, and machine learning.
Understanding of how an algorithm works is not required in order to produce predictions.	Understanding algorithms is as important as understanding experimental approaches.
Most computational techniques can be learned in a few hours of hands-on workshops.	How to execute a software might be learned rapidly, but understanding the theory behind a computational approach and gaining the experience essential to the correct interpretation of the relevance, meaning and reliability of predictions can be demanding and take a significant amount of time.

Bridging the gap

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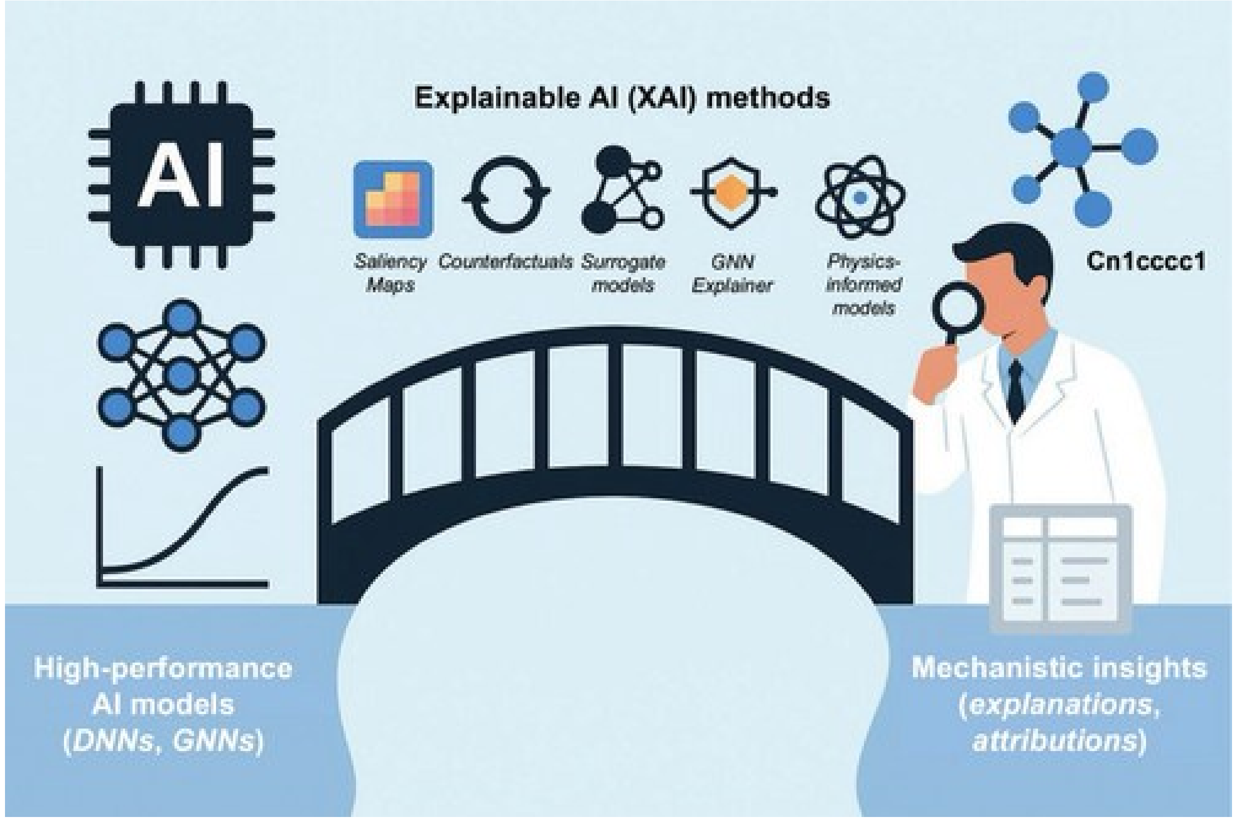
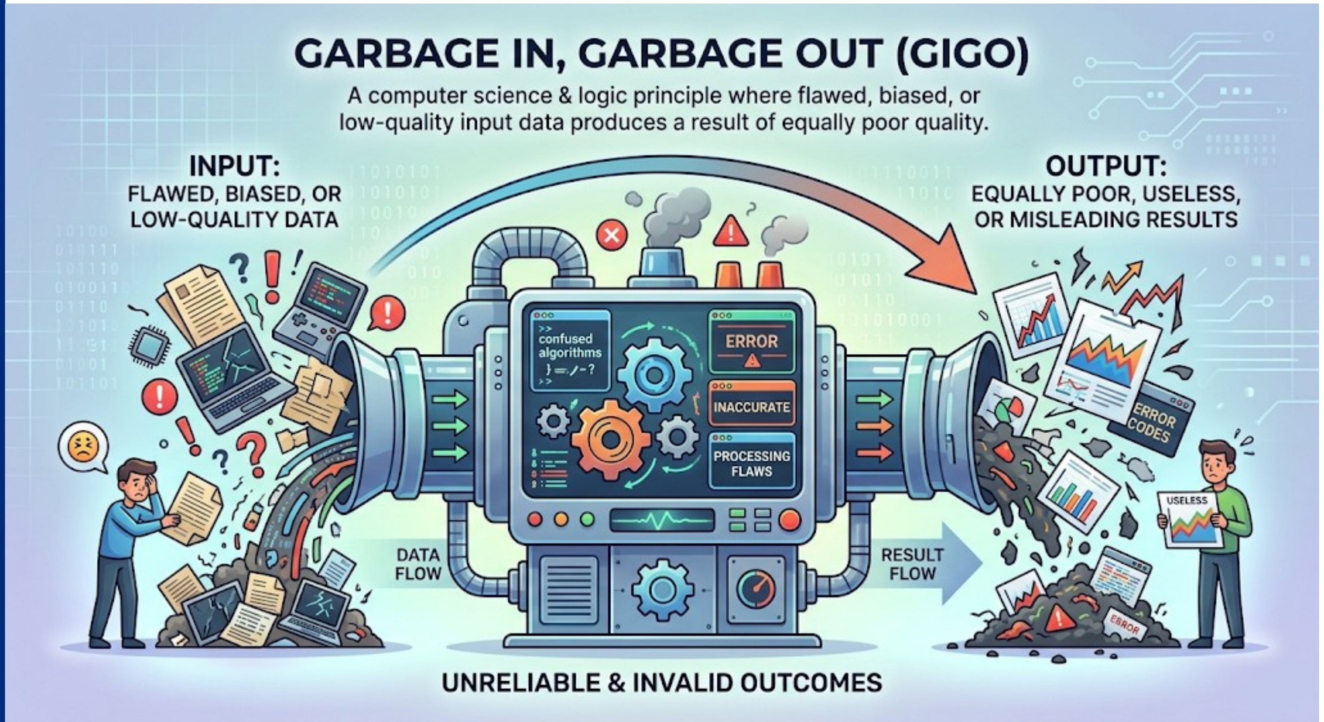
- Intense study of the literature.
- Problem-oriented rather than technique-oriented.
- Seek supervision or advice from experts.
- Avoid excessive use of buzzwords such as “artificial intelligence”, “machine learning” or “deep learning”
- Input data quality is of critical importance.



A. Lavecchia, “ Explainable Artificial Intelligence in Drug Discovery: Bridging Predictive Power and Mechanistic Insight,” Wiley Interdisciplinary Reviews: Computational Molecular Science 15, no. 5 (2025): e70049.

Reflections



- Bioinformatics, Chemoinformatics, and Biomedical Informatics are inherently interconnected in rational drug design.
- Avoid using artificial intelligence or machine learning just to follow trends. Use them with a clear methodological purpose.
- The success of predictions in CADD depends almost entirely on proper preliminary design and thorough data curation.
- Computing power does not replace scientific judgment or human reasoning in the interpretation of results.
- Never calculate what you don't understand.

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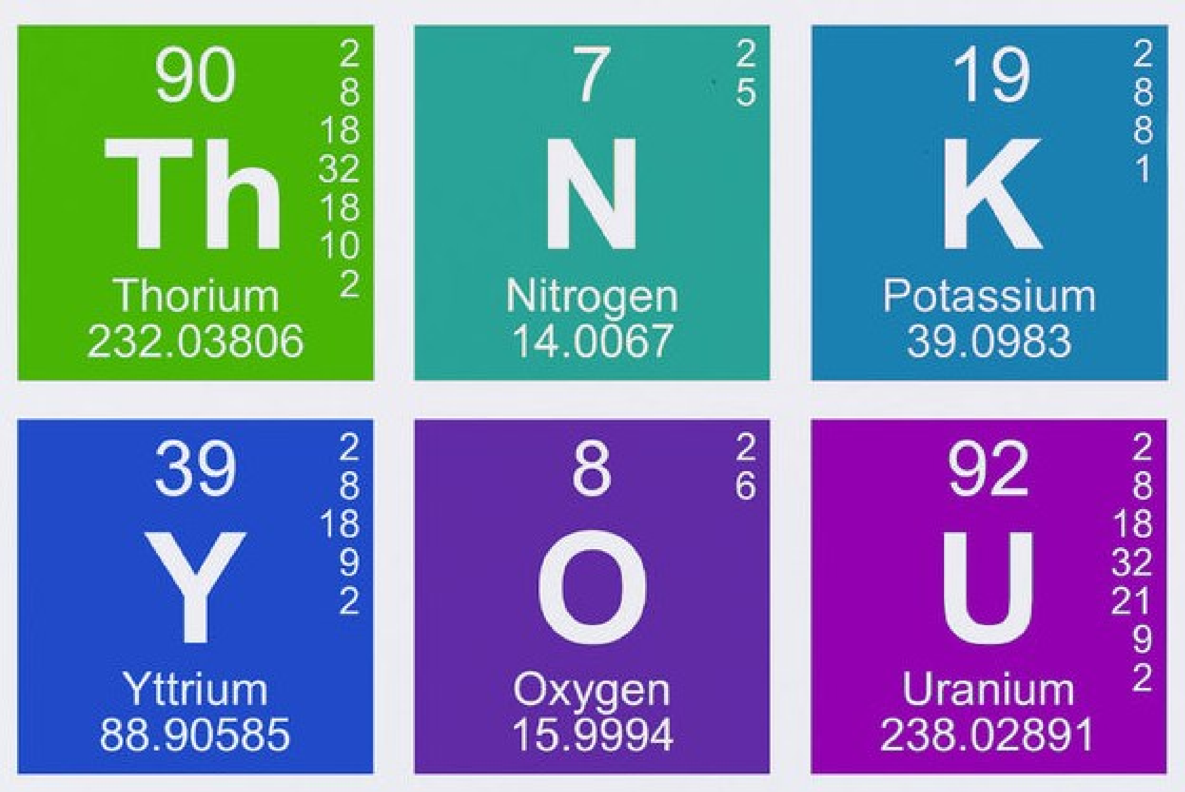

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